CARL CHRISTIAN JARQUE



Computer Engineering Graduate · Web Development & Workflow Automation · AI-Augmented Problem Solving

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PROFESSIONAL SUMMARY

Computer Engineering graduate building responsive web apps and automating manual workflows with React, TypeScript, and Firebase, leveraging Claude and ChatGPT daily to design and iterate faster. Comfortable across frontend development, UI/UX thinking, and structured logic design for digital workflows and AI systems.

EXPERIENCE

Freelance Frontend Developer — HS Technologies (Phils.), Inc. *06/2026 – 08/2026 · Contract / Remote*

- Continued development of the Digital Request & Approval System (formerly MRF Digitalization) system as an independent contractor after internship completion, engaged to finish and deploy it for company-wide employee use.
- Added an Admin Requests inbox, factory-based routing (Metal/Plastic), and a help-request flow distinguishing requesters from noters/conformers; expanded notifications to cover all workflow statuses.
- Coordinated with the backend developer and company stakeholders to resolve integration issues and prepare the system for full staff deployment.

QA Intern (Frontend Developer) — HS Technologies (Phils.), Inc. *03/2026 – 05/2026 · Rosario, Cavite*

- Developed Digital Request & Approval System (formerly MRF Digitalization) — a React + Firebase app replacing paper-based Modification Request Forms with a multi-step digital approval workflow, as one of two developers. Requests route through multiple approvers with real-time status updates via Firestore, and Firebase Auth ensures each approver only sees requests at their stage.
- Worked directly with QA department staff to understand their existing process and requirements. Integrated Cloudinary for media uploads and collaborated on core workflow architecture — component interaction logic and UI state transitions between approval stages — based on real usage feedback.
- Leveraged Claude throughout development to accelerate component scaffolding, debug logic errors, and refine UI copy — consistently reviewing and validating AI output against actual requirements.
- Built and maintained reusable UI components with React 18 and Tailwind CSS following component-based architecture and clean code standards; managed team codebase via Git with branch-based workflows.

SKILLS

Frontend Development: React 18 · TypeScript · JavaScript (ES6+) · HTML5 / CSS3 · JSX · Next.js (learning)

UI/UX & Styling: Tailwind CSS · MUI v5 · shadcn/ui · Responsive Design · Component Architecture

Logic & Flow Design: User flow mapping · Multi-step workflow design · Edge case handling · State transition logic · Scenario structuring

AI & Workflow Tools: Claude · ChatGPT · n8n · Prompt engineering · AI-assisted development · Generative AI literacy

Backend & Tools: Firebase (Auth + Firestore) · Socket.IO · Recharts / Chart.js · Cloudinary · Vite / npm · Git / GitHub · Netlify

Languages: English (proficient — written & spoken) · Filipino / Tagalog (native)

PROJECTS

Job Application Tracker | *n8n, Google Sheets, Gmail, OAuth2*

- Built two self-hosted n8n workflows that log each Google Form application to a Google Sheet and email a follow-up reminder two weeks later if the status is still "Applied". Documented at github.com/jarquecarl/n8n-job-application-tracker.
- Designed for reliability: upsert on a unique ID to avoid duplicate rows, a "due on or before today" check so missed runs catch up, retries on safe steps only, and explicit timezone handling.

PhiNex | *React, MUI, Socket.IO, Chart.js*

- Built the React 18 + MUI monitoring dashboard for a PYNQ-Z2 ARM-FPGA phishing detection system, with real-time threat data streamed via Socket.IO and visualized with Chart.js.

Fundo | *React, JavaScript, Recharts, Tailwind CSS, Vite*

- Personal expense tracker with category breakdowns, spending summaries, and interactive Recharts visualizations, plus a dark/light mode toggle.

Developer Portfolio v1 & v2 | *HTML/CSS → React, TypeScript, shadcn/ui, Netlify*

- Two-iteration portfolio — v1 in vanilla HTML/CSS, v2 rebuilt in React + TypeScript with shadcn/ui — both deployed on Netlify.

Maze Runner | *TypeScript*

- Browser-based game built in TypeScript — practiced strict typing, modular game-loop architecture, and multi-branch state management outside of traditional web app development.

EDUCATION

BS Computer Engineering — San Sebastian College Recoletos de Cavite · *2026 · Cavite City, Philippines*

Senior High School — STEM — San Sebastian College Recoletos de Cavite · *2019 · Cavite City, Philippines*

CERTIFICATIONS

- AI Learning Modules (12 hrs) — AIClassASEAN / ASEAN Foundation
- AI Ready ASEAN Seminar: "Hello, AI World" — AI Ready ASEAN / ASEAN Foundation
- Int'l Experts Sharing Meeting: Automation & Intelligent Control in Equipment Manufacturing — SEAMEO TED
- Int'l Perspectives Webinar: Modular Solar Chimney Dryer — Go Study Global Education
- Hour of Code (AI Ready ASEAN Programme) — Break The Fake Movement x ASEAN Foundation